TECHNICAL REPORT

Contextro: A Pure-Rust MCP Server for Token-Efficient Repository Discovery

A controlled 1,000-task evaluation of deterministic repository-discovery workflows on a production TypeScript repository

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This manuscript supersedes the earlier pre-Rust Contextro paper. All headline results here are backed by the current Rust benchmark harness and the machine-readable artifacts bundled in `docs/publication/`.

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ABSTRACT

Repository discovery still dominates the prompt budget of many MCP-based coding workflows. This paper evaluates the current Rust-era version of Contextro, a pure-Rust local MCP server, with a deterministic 1,000-task paired study on a private production repository named `platform`. The benchmark target contains 10,889 tracked files; Contextro indexed 8,498 source files into 11,534 symbols and 11,534 chunks. Against a stronger scripted local baseline built from exact `git grep` plus bounded file reads, Contextro reduced total token volume from 1,068,884 to 94,034 (91.2%), reduced median handler latency from 202.969 ms to 0.037 ms, reduced mean tool calls from 3.098 to 1.000, and eliminated file reads in the study harness. Contextro completed all 1,000 tasks; the baseline completed 999. Because the current TypeScript and JavaScript parser path emits zero call relationships on this repository, graph-dependent tasks were explicitly excluded. The resulting claim is therefore narrow and honest: on this repository, the current pure-Rust Contextro implementation materially reduces retrieval overhead for deterministic discovery and symbol-navigation tasks.

Keywords: Model Context Protocol, repository discovery, Rust, code intelligence, BM25 retrieval, benchmark design

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1. Introduction

Most coding agents still locate code the expensive way. They grep broadly, open several files, inspect mostly irrelevant windows, and only then recover the symbol or file they needed. That workflow wastes tokens because it sends exploration traces rather than direct answers.

Contextro is built to replace that exploration trace with targeted repository intelligence. It exposes indexed search, symbol lookup, code operations, history lookup, and memory through a local MCP server so a client can ask for a definition, a ranked exact-search result, or a file-local symbol list directly.

The earlier Contextro paper is no longer the right description of the system. The current implementation is Rust end to end, and the benchmark in this paper is produced by a real Rust study harness that invokes the current retrieval handlers directly. No Python runtime and no legacy `ctx_fast` path remain in the evaluated system.

1. We report the first controlled 1,000-task benchmark for the current pure-Rust Contextro implementation on the private `platform` repository.
2. We compare Contextro against a stronger scripted local baseline based on exact `git grep` and bounded file reads, not against a simulator.
3. We state capability limits explicitly: graph-dependent tasks are excluded because the fresh TypeScript-heavy index contained zero call relationships.

2. Related Work

This work sits at the intersection of code search, retrieval-augmented generation, and prompt-budget management. BM25-style lexical retrieval remains a strong baseline for identifier-heavy search, while structured retrieval and code graphs are increasingly used to keep only relevant evidence in the context window. Our contribution is narrower than a general code-generation paper: we measure repository-discovery cost inside MCP-mediated workflows and ask how much of that cost disappears when the client can query a local repository-intelligence layer instead of traversing files manually.

3. Current System

Contextro is now implemented as a pure-Rust local MCP server. The workspace separates indexing, engines, parsing, memory, git helpers, tool handlers, and the server binary across focused crates, but the operational surface is still a single local binary. For this study, the important detail is architectural continuity: the benchmark harness reuses the same Rust retrieval paths that the server exposes, so the benchmark and the product measure the same code path rather than two different implementations.

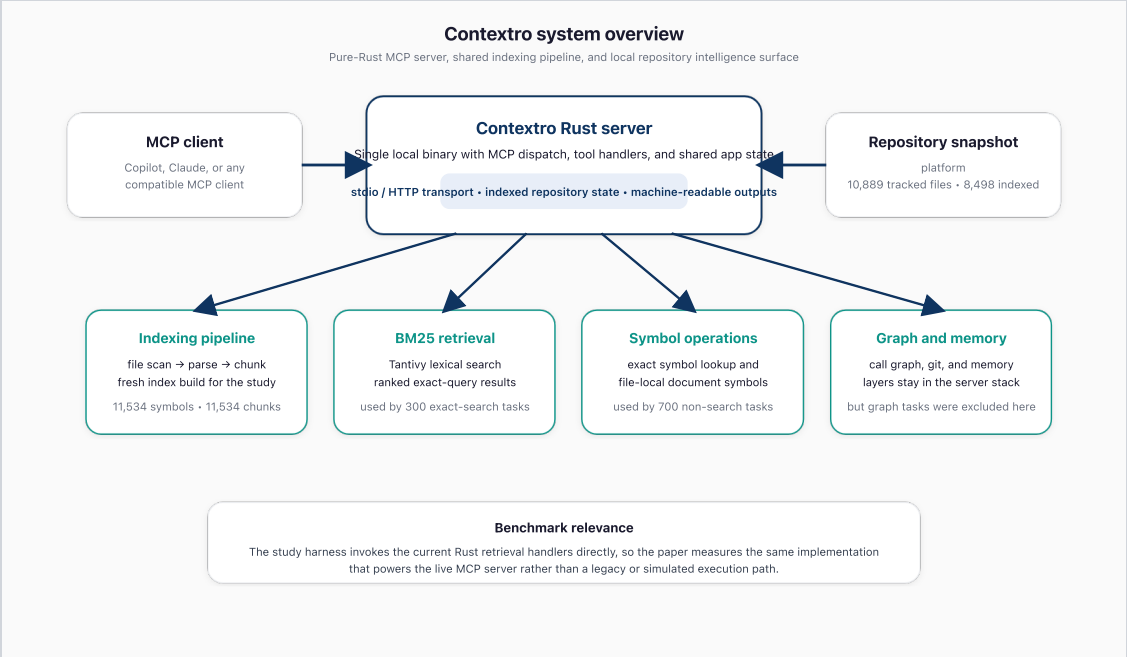

Figure 1. Current Contextro system overview. The benchmark exercises the same Rust indexing and retrieval stack that powers the server runtime.

3.1. Implementation Snapshot

Table 1. Current Rust implementation snapshot.

Layer	Current implementation	Role in this study
Runtime and transport	Rust binaries in `crates/contextro-server` with `rmcp`, `tokio`, and `axum`	Expose the local MCP server and the `contextro-study` benchmark harness
Indexing	`contextro-indexing` pipeline over supported source files	Built a fresh 8,498-file index with 11,534 symbols and 11,534 chunks
Lexical retrieval	In-memory Tantivy BM25 index	Powers the 300 exact-search tasks in the treatment arm
Symbol lookup	In-memory Rust graph and name indexes	Powers exact `find_symbol` and grouped lookup tasks
Code operations	Rust `code` handlers plus `TreeSitterParser` interface	Powers `lookup_symbols` and `get_document_symbols`
Current JS/TS limitation	The current parser path does not populate `calls` for TypeScript or JavaScript symbols on this repo	Produced 0 graph relationships, so graph-dependent tasks were excluded

3.2. Evaluated Tool Surface

Table 2. Tool surface exercised by the 1,000-task study.

Study category	Treatment tool path	Tasks	Deterministic success condition
Symbol discovery	`find_symbol(name, exact=true)`	300	The expected file path appears in the returned symbol set
Exact search	`search(query, mode="bm25", limit=5)`	300	The expected file path appears in the ranked results
Batch lookup	`code(operation="lookup_symbols")`	200	All expected file paths for the three-symbol batch appear
Document symbols	`code(operation="get_document_symbols")`	200	All expected symbol names appear in the file-level symbol list
Excluded graph tasks	`find_callers`, `find_callees`, `impact`, relationship-rich `explain`	0	Explicitly excluded because the fresh index contained zero graph relationships

The study harness uses the current Rust handlers directly. An older JavaScript experiment script in the repository is only a simulator and is not used for any headline claim in this paper.

4. Benchmark Methodology

The study follows a paired, deterministic, two-arm design. Every task is executed once with a stronger scripted local baseline and once with Contextro against the same repository snapshot. The task generator draws the inventory from indexed ground truth rather than from free-form human prompts, so correctness can be scored mechanically.

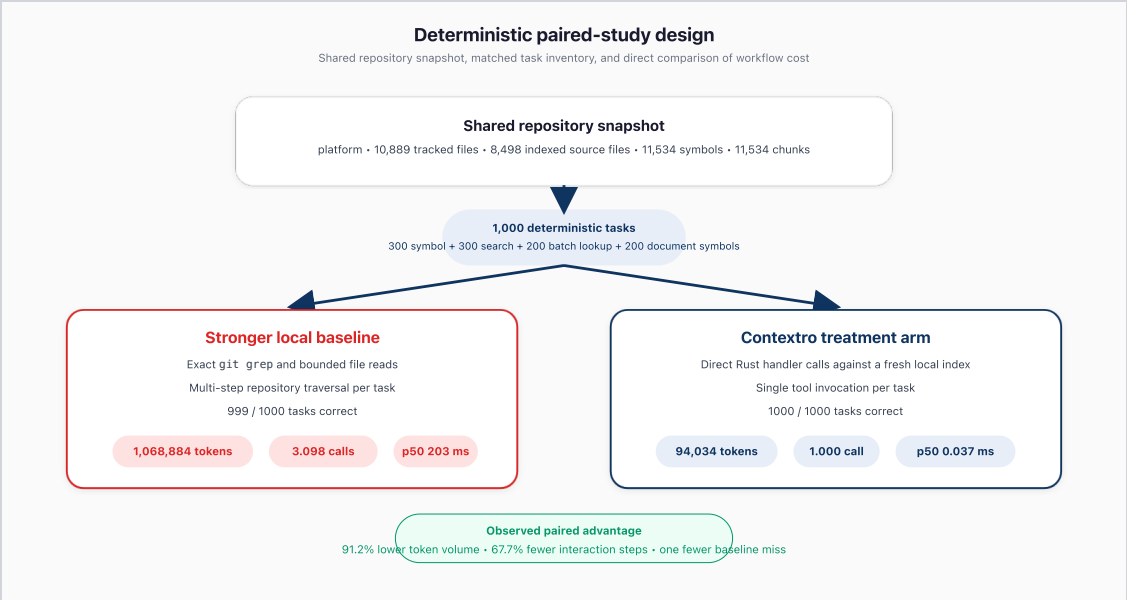

Figure 2. Deterministic paired-study design. The same 1,000-task inventory is executed in both arms against the same repository snapshot.

4.1. Study Design

The stronger local baseline uses exact `git grep` plus bounded file windows. Symbol-discovery tasks read up to two candidate files after a grep hit; exact-search tasks read up to three; batch-lookup tasks run three exact greps and read one primary file window for each symbol; document-symbol tasks open a single file excerpt directly. The treatment arm performs exactly one Contextro tool call per task against a pre-indexed repository state.

The task inventory contains 300 symbol-discovery tasks, 300 exact-search tasks, 200 batch-lookup tasks, and 200 document-symbol tasks. Batch and document-symbol tasks are intentionally stricter than single-file lookups because all expected files or symbols must appear for the task to count as correct.

4.2. Repository Under Test

Table 3. Repository under test and benchmark inventory.

Property	Value
Benchmark target	Private production repository `platform`
Tracked files	10,889
Dominant tracked extensions	5,545 `.ts` files and 2,905 `.tsx` files
Indexed source files	8,498
Extracted symbols / chunks	11,534 / 11,534
Graph nodes / relationships	11,534 / 0
Task distribution	300 symbol discovery, 300 exact search, 200 batch lookup, 200 document symbols
Fresh index build	0.45 s
BM25 build	34.3 ms
Graph build	13.5 ms
Benchmark host	Apple M4, 16 GB RAM, macOS 26.4.1

4.3. Metrics and Deterministic Scoring

Table 4. Metrics and scoring rules.

Measurement	Definition	Purpose
Success	Expected file or symbol strings must appear in the rendered output; batch and document tasks require all expected items	Removes subjective judging and keeps the benchmark reproducible
Token estimate	`tiktoken-rs` with `cl100k_base` applied to each arm's rendered output	Provides an apples-to-apples context-cost comparison
Handler latency	Wall-clock milliseconds around each scripted arm call	Measures retrieval overhead inside the harness on a pre-indexed repository
Tool calls	Count of grep and read operations in the baseline versus one MCP call in the treatment arm	Captures workflow complexity
Files read	Count of explicit file windows opened by the scripted baseline	Measures raw code traversal cost
Excluded capabilities	Graph-dependent tasks are removed when the fresh index has zero relationships	Prevents us from claiming performance for surfaces the current parser did not support on this repo

This benchmark is intentionally not an autonomous coding benchmark. It studies retrieval, symbol navigation, and file-local structure under deterministic scoring. That choice narrows the claim, but it also makes the result reproducible and automatically checkable.

5. Results

5.1. Headline Outcomes

The full 1,000-task study produced a consistent result: Contextro reduced the amount of text emitted by the discovery workflow and did so without sacrificing correctness. Total token volume fell from 1,068,884 in the stronger local baseline to 94,034 in Contextro, a 91.2% reduction. The baseline succeeded on 999 tasks; Contextro succeeded on all 1,000.

Interaction cost also collapsed. The scripted baseline averaged 3.098 tool invocations and 1.898 file reads per task, while the Contextro arm issued exactly one tool call and read no files inside the study harness. Median handler latency fell from 202.969 ms to 0.037 ms, though that latency should be interpreted carefully as direct handler timing on a pre-indexed local repository rather than end-to-end agent wall clock.

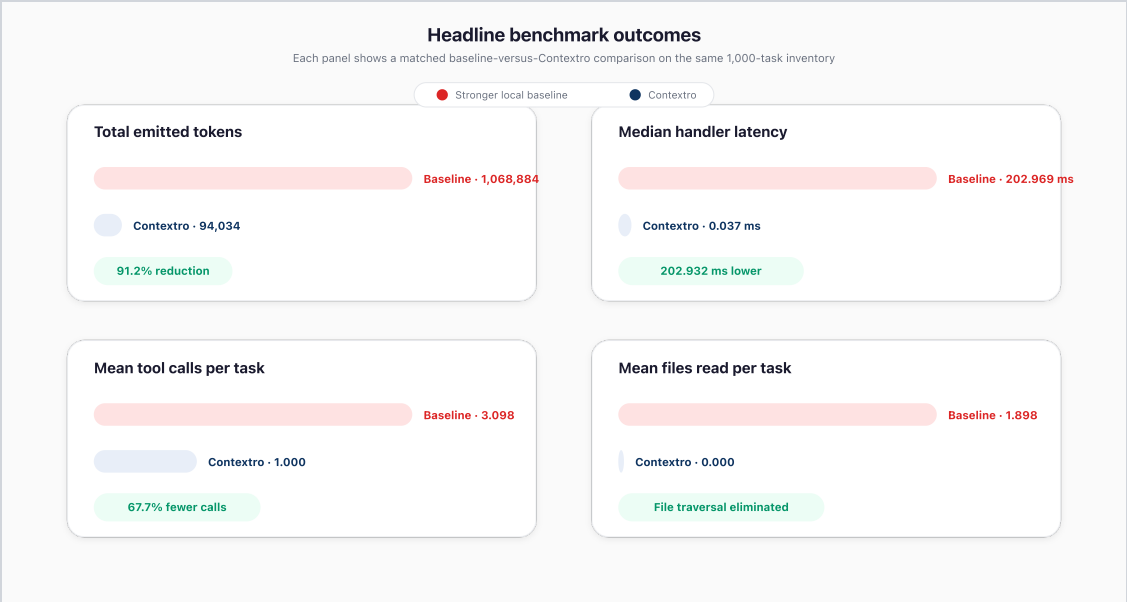

Figure 3. Headline benchmark outcomes. Contextro cuts token volume and interaction cost sharply while matching or exceeding baseline correctness.

Table 5. Headline outcomes from the 1,000-task study.

Metric	Stronger local baseline	Contextro	Observed change
Success rate	99.9% (999 / 1000)	100.0% (1000 / 1000)	Contextro completed every task in the inventory
Total tokens	1,068,884	94,034	91.2% reduction
Mean tokens per task	1,068.9	94.0	91.2% reduction
Median handler latency per task	202.969 ms	0.037 ms	Direct local handler timing dropped by 202.932 ms
P95 handler latency per task	624.947 ms	0.094 ms	The long tail remained entirely on the baseline side
Mean tool calls per task	3.098	1.000	67.7% fewer interaction steps
Mean files read per task	1.898	0.000	Explicit file reads were eliminated in the treatment arm

5.2. Results by Task Category

All four categories favored Contextro on token volume. The largest reduction appeared in symbol discovery (94.5%), where the treatment arm can usually answer the query with a single short symbol record. The smallest reduction appeared in document-symbol tasks (86.8%), which is expected because those tasks require the system to emit a list of file-local symbol names rather than a single location.

Latency splits by category are also informative. The biggest latency gain appears in batch lookup because the baseline must perform three exact greps and three file reads. Document-symbol tasks are the exception: the baseline is already close to direct file access, so latency is roughly equal there even though Contextro still reduces output size materially.

Table 6. Category-level outcomes.

Category	Tasks	Success (baseline / Contextro)	Tokens (baseline / Contextro)	Reduction	Median handler latency (baseline / Contextro)
Symbol discovery	300	100.0% / 100.0%	239,344 / 13,174	94.5%	204.238 ms / 0.016 ms
Exact search	300	99.7% / 100.0%	256,453 / 20,986	91.8%	202.114 ms / 0.066 ms
Batch lookup	200	100.0% / 100.0%	278,959 / 20,947	92.5%	612.560 ms / 0.029 ms
Document symbols	200	100.0% / 100.0%	294,128 / 38,927	86.8%	0.046 ms / 0.045 ms

5.3. Failure Analysis

The stronger local baseline missed exactly one task: `search_0553`, an exact-search query for `DataGrid`. The baseline recovered the symbol name, but its grep-plus-window path did not surface the expected target file `apps/app/src/features/data-grid-live/components/data-grid/data-grid.tsx`. Contextro returned the expected file in the ranked result set. This is a small miss, but it is useful because it shows where repeated identifier names still make a multi-step grep workflow brittle even under deterministic scripting.

The more important qualitative result is the absence of silent degradation in the treatment arm. Contextro was correct on every task category that the harness exercised. That does not prove broad coding superiority, but it does support the narrower claim that the current Rust implementation can answer deterministic discovery tasks compactly and reliably on this repository.

6. Discussion

The central result is a workflow result, not a marketing one. When the retrieval layer can answer a discovery question directly, the agent no longer needs to spend most of its context budget on exploratory file traversal. The 91.2% token reduction in this study is therefore best understood as a reduction in exploration waste: fewer grep traces, fewer file windows, and fewer intermediate steps.

The pure-Rust transition matters for the credibility of that claim. The current server, the current handlers, and the current benchmark harness all live in the same Rust codebase. The numbers in this paper do not come from the earlier Python-era architecture and do not depend on the older JavaScript simulator. They describe the current MCP as it exists now.

The category breakdown also keeps the result honest. Document-symbol tasks do not show dramatic latency gains because the baseline is already close to direct file access. Contextro still helps there because the response is more compact, but not every category should be expected to improve in the same way.

7. Limitations

This paper has four important limitations. First, it studies a single private repository, so the result should be read as strong evidence on `platform`, not as a universal repository claim. Second, the benchmark uses a pre-indexed local repository and measures direct handler latency, not full end-to-end agent runtime with model reasoning and transport overhead. Third, the task inventory is deterministic and retrieval-oriented rather than a full autonomous coding benchmark.

Fourth, and most importantly, the current TypeScript and JavaScript parser path produced zero call relationships on this repository. That forced us to exclude `find_callers`, `find_callees`, `impact`, and relationship-rich `explain` tasks from the study. The paper therefore should not be read as evidence that the current system already delivers graph-quality answers on TypeScript-heavy repositories. Fixing that parser limitation is the most important technical follow-up after this benchmark.

The benchmark also favors exact identifier retrieval. The 300 exact-search tasks use `search(mode="bm25")` because that path admits deterministic scoring from indexed ground truth. That makes the result clean and reproducible, but it leaves open a separate question about free-form natural-language search quality, which deserves its own study.

8. Availability and Reproducibility

The primary artifact bundle for this paper lives in `docs/publication/`. The four new platform-study JSON files are the machine-readable evidence for every headline number in the manuscript. Older paired-study and public-proxy artifacts remain in the directory for historical reference, but they are not the evidence base for the claims in this paper.

8.1. Artifact Package

Table 7. Primary publication artifacts for the Rust-era paper.

Artifact	Description
`contextro-paper.html`	Printable manuscript source for the current Rust-era paper
`platform-1000-study-config.json`	Benchmark configuration, tokenizer metadata, and index snapshot
`platform-1000-study-tasks.json`	Full deterministic 1,000-task inventory
`platform-1000-study-results.json`	Per-task, per-arm outcomes for all 2,000 executions
`platform-1000-study-summary.json`	Aggregate results cited throughout this paper

8.2. Rerun Command

```
cd crates
cargo run --release -p contextro --bin contextro-study -- \
  --codebase /Users/japneetkalkat/platform \
  --output-dir
/Users/japneetkalkat/conductor/workspaces/contextro/zurich/docs/publicati
on \
  --tasks 1000
```

8.3. Verification Environment

Table 8. Verification environment.

Field	Value
CPU	Apple M4
Memory	16 GB RAM
Operating system	macOS 26.4.1
Tokenizer	`tiktoken-rs` with `cl100k_base`

9. Conclusion

On the private `platform` repository, the current pure-Rust Contextro implementation makes repository discovery substantially cheaper. In a deterministic 1,000-task study, it cut token volume by 91.2%, collapsed the number of interaction steps to one call per task, and completed every task in the inventory. Those are meaningful retrieval wins, even though the benchmark does not attempt to measure full autonomous coding quality.

The next step is clear. The benchmark surfaced a real parser limitation on TypeScript-heavy repositories: graph relationships are currently absent, so graph-native tasks cannot yet be evaluated honestly. Solving that limitation should be the priority before claiming broad impact-analysis or caller-tracing performance on JavaScript and TypeScript codebases.

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Appendix A. Experiment Configuration

Table 9 collects the configuration values that define the benchmark. These fields are duplicated in `platform-1000-study-config.json`.

Table 9. Experiment configuration.

Field	Value
Requested tasks	1,000
Generated tasks	1,000
Task families	300 symbol discovery, 300 exact search, 200 batch lookup, 200 document symbols
Tracked files	10,889
Indexed files	8,498
Total symbols / chunks	11,534 / 11,534
Graph relationships	0
Index build times	0.45 s indexing, 34.3 ms BM25 build, 13.5 ms graph build
Baseline strategy	Exact `git grep` plus bounded file windows
Treatment tools	`find_symbol`, `search(mode="bm25")`, `code.lookup_symbols`, `code.get_document_symbols`

Appendix B. Tool Contracts Used in the Study

This appendix summarizes only the four tool paths exercised in the 1,000-task benchmark. The benchmark interacts with the handlers directly, but the structures below match the externally visible MCP surface.

Table 10. Tool contracts used in the study.

Tool	Inputs used here	Primary output fields checked	Role
`find_symbol`	`name`, `exact=true`	`symbols[].file`, `symbols[].name`, `symbols[].line`	Exact definition discovery
`search`	`query`, `limit=5`, `mode="bm25"`	`results[].file`, `results[].name`, `results[].score`	Ranked exact-identifier retrieval
`code(operation="lookup_symbols")`	`symbols` list	`symbols[].file`, `symbols[].name`, `symbols[].line`	Grouped three-symbol lookup
`code(operation="get_document_symbols")`	`file_path`	`symbols[].name`, `symbols[].line`, `symbols[].signature`	File-local symbol inventory

Appendix C. Category-Level Interaction Costs

Table 11 reports interaction-heavy details aggregated from `platform-1000-study-results.json`.

Table 11. Per-category interaction costs.

Category	Baseline mean tool calls	Contextro mean tool calls	Baseline mean files read	Contextro mean files read	Baseline p95 latency	Contextro p95 latency
Symbol discovery	2.763	1.000	1.763	0.000	228.248 ms	0.031 ms
Exact search	2.897	1.000	1.897	0.000	218.685 ms	0.112 ms
Batch lookup	6.000	1.000	3.000	0.000	663.968 ms	0.058 ms
Document symbols	1.000	1.000	1.000	0.000	0.107 ms	0.131 ms